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Assignment: Implementing Polymorphism

Objective

To understand and implement polymorphism in Java by using method overriding and superclass references with previously created inheritance hierarchies.

Problem Statement

Using the same class hierarchies implemented in the *Inheritance in Java* assignment, you are required to demonstrate polymorphism in Java.

This assignment focuses on:

- Method overriding
- Superclass reference pointing to subclass objects
- polymorphism (generalized reference algorithm)

Student Tasks

1. **Add a Common Method in Each Superclass**

   Add at least one common method in every superclass.
   Examples (you may choose appropriate names based on hierarchy):

   - `calSalary()`
   - `brake()`
   - `calculateArea()`
   - `play()`
2. **Override the Method in All Subclasses**
   - Override the same method in each corresponding subclass
   - Each subclass must provide its own implementation
   - Method name and signature must remain exactly the same
3. **Use Superclass References**
   - Create reference variables of the superclass
   - Assign them objects of different subclasses
4. **Demonstrate Polymorphism**
   - Invoke the overridden method using the superclass reference
   - Observe different outputs based on the object type at runtime

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Implementation Guidelines

• Use the same hierarchies from the Inheritance assignment
• Do not create new hierarchies for this assignment
• Method names must be the same in superclass and subclasses
• Method signatures must match exactly
• Overriding must follow Java rules

Main Method Requirements

- Create at least one superclass reference
- Assign it to objects of different subclasses
- Call the overridden method using the superclass reference
- Display output to show different behavior for each subclass

Important Rules

• Follow proper Java naming conventions
• Use clean and readable code
• Do not use method overloading for this assignment

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